When Interest Rates Ignored Inflation

Gibson's paradox in British data, 1729-2016

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Abstract

While the pound was convertible into gold, British long-term interest rates moved with the level of prices and were almost unrelated to inflation, the regularity Keynes called Gibson's paradox. In the Bank of England's annual data, the consol yield over 1822-1913 has a correlation of 0.45 with the wholesale price level around a linear trend and 0.47 around a cubic trend, 0.61 when both series are measured against their own 31-year moving averages, and 0.06 with inflation over the previous five years. A high price level did not forecast inflation over the following decade, so expected inflation does not explain the comovement. Under paper money the pattern reverses. During the Restriction of 1797-1821, when sterling was inconvertible, the correlation with the price level is −0.14 and with inflation 0.41, and since 1973 the correlation with inflation is 0.85. In the Jordà-Schularick-Taylor data the median correlation with inflation is 0.03 across 17 gold-standard countries and 0.83 after 1973, positive in all 18. The evidence fits Barsky and Summers (1988), in whose model a fixed gold price makes the price level the inverse of gold's real value, which falls when real interest rates rise.

1 The paradox

Gibson (1923) observed that British long-term yields and the price level moved together for over a century. Fisher's theory of interest says that nominal rates should move with expected inflation, which is the rate of change of prices and not their level, so Keynes (1930) called the regularity a paradox and one of the most firmly established facts in quantitative economics. Barsky and Summers (1988) offered an explanation that ties it to the monetary regime. Under a gold standard the nominal price of gold is fixed, so the price level is the inverse of the real price of gold. Gold is a durable asset, and when real interest rates rise its real value falls, which raises the price level. With prices roughly stable on average, nominal yields are close to real yields, and the two move together. The explanation predicts that the paradox should disappear when money is not convertible into gold and the price level is set by the monetary authority.

British data allow two tests of that prediction. The Bank of England's "A millennium of macroeconomic data" (Thomas and Dimsdale 2017) has annual consol yields from 1703 and wholesale prices from 1661. Sterling was convertible into gold, de facto from 1717 and formally after convertibility resumed in 1821, except during the Restriction of 1797-1821, when the Bank of England suspended convertibility during the wars with France, and it left gold for good in 1931. Within the metallic era the Restriction is a 25-year spell of paper money, and the modern period since 1973 is a longer one. We use the consol (long-term government) yield and the log wholesale price index, which is the price measure Gibson used, and measure inflation as the average change in log wholesale prices over the previous five years. For other countries we use the Jordà-Schularick-Taylor Macrohistory Database (Jordà, Schularick and Taylor 2017), release 6, with long-term government bond yields and consumer prices for 18 countries from 1870 to 2020.

2 Results

The paradox is a feature of the classical gold standard (Table 1). Over 1822-1913 the consol yield has a correlation of 0.45 with the price level around a linear trend, 0.47 around a cubic trend and 0.61 when both series are measured as deviations from their own 31-year moving averages. Its correlation with inflation over the previous five years is 0.06. Prices in this period went through several long swings, rising to 1873, falling to 1896 and rising again to 1913, as Figure 1 shows, so the result does not rest on a single cycle. Expected inflation cannot account for it, since inflation averaged −0.4 percent a year and the price level relative to its average over the previous 30 years did not forecast inflation over the following ten years (slope 0.025, Newey-West standard error 0.042). The 18th century under the de facto gold standard shows the same relation more weakly, with correlations of 0.30 to 0.58, and the yield is also correlated with inflation (0.36).

Under paper money the price-level relation changes sign. During the Restriction the correlation with the detrended price level is −0.14 around a linear trend and −0.33 with both series measured against

Regime	Price level, detrended by linear trend	Price level, detrended by cubic trend	Both series, 31-year average	Inflation	Years
Gold (de facto), 1729-1796	0.35	0.30	0.58	0.36	68
Paper pound, 1797-1821	−0.14	−0.23	−0.33	0.41	25
Gold, 1822-1913	0.45	0.47	0.61	0.06	92
Interwar, 1920-1938	0.12	0.23	0.84	0.19	19
Bretton Woods, 1950-1971	0.08	0.00	0.18	−0.54	22
Fiat, 1973-2016	−0.02	0.08	−0.29	0.85	44

Table 1: Correlation of the annual British consol yield with the log wholesale price level and with inflation, by monetary regime. In the first two columns the log price level is detrended by a linear or cubic trend fitted within each regime and the yield is used in levels. In the third column both the yield and the log price level are measured as deviations from their own 31-year centered moving averages, computed on the full series, as in Figure 1, so the fiat row for that column ends in 2001 (29 years). Inflation is the average annual change in log wholesale prices over the previous five years. Data from the Bank of England, A millennium of macroeconomic data, version 3.1.

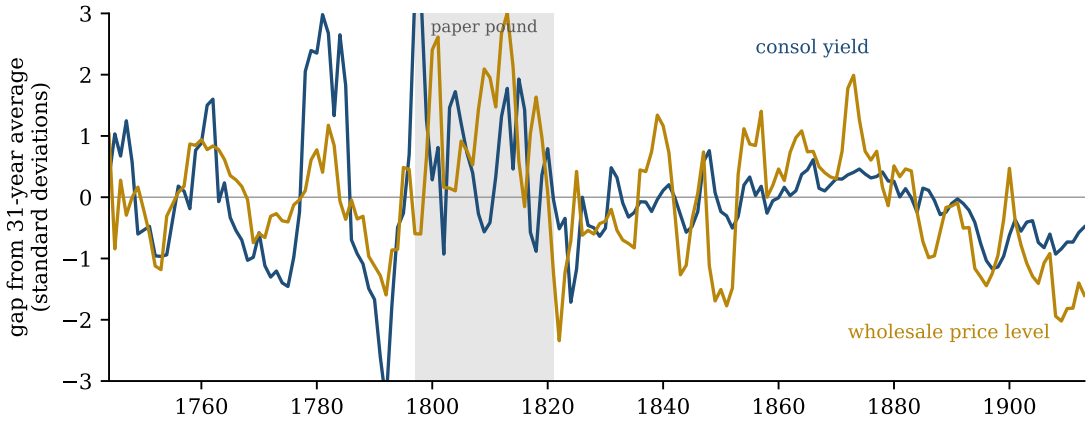


Figure 1: British consol yield and log wholesale price level, each measured as the deviation from its own 31-year centered moving average and standardized over 1744-1913. The shaded years are the Restriction of 1797-1821, when sterling was inconvertible.

their moving averages, and yields move with inflation instead, with a correlation of 0.41. Since 1973 the correlation with inflation is 0.85. Inflation averaged 5.3 percent a year over 1973-2016 with a standard deviation of 4.7 points, compared with 2.7 points under gold, and the consol yield followed it.

Regime	Price level, detrended by linear trend	Price level, detrended by quadratic trend	Both series, 31-year average	Inflation
Gold, 1875-1913	0.62 (0.94)	0.09	0.57 (0.82)	0.03 (0.59)
Interwar, 1920-1938	−0.01 (0.50)	0.01	0.77 (1.00)	0.44 (0.94)
Bretton Woods, 1950-1971	0.20 (0.78)	0.08	0.32 (0.80)	−0.05 (0.44)
Fiat, 1973-2020	0.31 (0.89)	0.26	0.52 (0.94)	0.83 (1.00)

Table 2: Median across countries of the correlation of the long-term government bond yield with the log consumer price level and with inflation over the previous five years, with the share of countries with a positive correlation in parentheses. The linear, quadratic and inflation columns use only years in which five-year inflation is defined, so the gold period starts in 1875, and countries with at least 12 such years in a period: 17 in the gold period and 18 thereafter. In the 31-year average column both the yield and the log price level are deviations from their own 31-year centered moving averages, which requires 15 years of data on each side, and countries need at least 12 years of these deviations in a period: 17 countries over 1885-1913, 12 in 1920-1938, 15 in 1950-1971 and 18 over 1973-2005. Data from the Jordà-Schularick-Taylor Macrohistory Database, release 6.

The cross-country data agree on inflation and are weaker on the price level (Table 2). Under the gold standard the median correlation of long-term yields with inflation across 17 countries is 0.03, and after 1973 it is 0.83, positive in every country. The gold-standard correlation with the price level is 0.62 around a linear trend and positive in 16 of 17 countries, with the strongest values in Britain (0.83) and the United States (0.79). The 1875-1913 window, however, contains essentially one swing in prices, down to 1896 and up to 1913, which all gold countries shared, and a quadratic trend removes it. The longer British series is needed to show that the relation repeats.

3 Scope

Under fiat money the price-level measures are hard to read. The price level since 1973 inherits the shape of inflation, rising fast in the 1970s and slowly after the 1980s disinflation, so any detrended price level contains a hump that lines up with the hump in interest rates. This probably explains why the fiat-era price-level correlations are positive in the cross-country data (0.26 to 0.52 across the three measures) but near zero or negative in Britain. The contrast that survives every measure is the correlation with inflation, between 0.03 and 0.06 under the gold standard and above 0.8 since 1973. The Restriction coincided with the French wars, so war finance, as well as the inconvertible pound, may explain why yields followed inflation in those years. The correlations describe comovement and do not separate the Barsky-Summers channel from other explanations that depend on the monetary regime.

References

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